

Project Ideas

1. Real-Time Emergency & Public Safety Pipelines

These projects leverage your background in crisis management to build "high-uptime" software that provides immediate value in stressful situations.

- Real-Time Fire Perimeter & Population Exposure Pipeline: Build an automated system that polls wildfire data from APIs (like InciWeb or IRWIN), buffers the affected areas, and intersects them with Census population data to calculate threaten residents. This demonstrates your ability to handle automated data ingestion and spatial analysis in a context that mirrors your emergency service roots.
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- Weather-Triggered Asset Alerting System: Develop a pipeline that monitors the NWS API for active alerts and sends automated summaries to stakeholders identifying specific warehouses or critical infrastructure in danger zones. This showcases infrastructure thinking and the ability to build tools for high-stakes logistics or insurance sectors.
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2. Full-Stack Civic Engagement & Reporting Platforms

Your experience with de-escalation and stakeholder management makes you a strong candidate for building tools that bridge the gap between citizens and authorities.

- Community Infrastructure Reporting Platform: Model a project after Viralkan, a full-stack platform where citizens report road damage. This involves building a Node.js or Hono backend, implementing Google OAuth for secure login, and utilizing PostGIS for geographic data storage.
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- No-Auth Ingestion Validation Pipeline: Create a reporting endpoint that allows public participation without mandatory registration. Use Zod schemas to validate incoming JSON payloads and rate-limiting middleware to prevent denial-of-service (DoS) attacks. This project proves you can manage security and "frictionless" user experiences simultaneously.
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3. Security Tooling & Cryptographic Systems

These projects align with your interest in "taking things apart" and coding to solve puzzles.

- Python-Based Security "Swiss Army Knife": Develop custom automation scripts for reconnaissance, such as a tool that performs subdomain discovery or scans for open ports. This demonstrates the "builder" mindset over the "user" mindset.

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- Cryptographic Password Manager: Build a CLI or GUI tool in Python that uses a cryptography library to encrypt and securely store credentials. This project serves as a "big deal" in interviews because it showcases your understanding of data persistence and core security principles.

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4. Hardware & Software Testing (IoT focus)

Given your interest in hardware testing, projects that bridge the physical-digital divide are ideal.

- IoT Security Vulnerability Lab: Set up a lab to test connected devices. Analyze firmware for unencrypted communication or hardcoded credentials and document a hardening guide for the device. This directly prepares you for specialized roles in IoT security.

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- Mobile App Decompilation & API Testing: Use Android Studio and debugging frameworks to test mobile applications for common vulnerabilities like insecure data storage. This highlights your interest in the "tester" role and your ability to use professional debugging tools.

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5. Automated Quality Assurance & Site Reliability

These projects demonstrate your ability to reduce manual backlogs and maintain system reliability—qualities essential for an SRE or QA Engineer.

- Automated Geocoding & Data Quality Pipeline: Build an ETL (Extract, Transform, Load) pipeline that takes messy address lists, geocodes them via API, and scores the results for match quality to flag records needing human review. This tells hiring managers you can automate "grunt work" to increase team efficiency on day one.

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- Docker-Containerized PostGIS Analysis Environment: Create a Docker Compose setup that spins up a PostGIS database and a Jupyter notebook server, pre-configured for spatial SQL queries. This reproducible environment proves you can set up the infrastructure substrate your code runs on

